By examining decision logs, teams could view the tangible outcomes of these interactions. AI tools proved helpful in summarizing and analyzing this data. Instead of manually reviewing thousands of authorization decisions, engineers could ask questions such as:

- “Who has accessed the Q3 plan in the last 30 days?”
- “Which policy most often permits access to customer documents?”
- “Are there principals who can access documents we did not expect them to see?”

Because these questions were answered on the basis of observed decisions rather than inferred intent, they often revealed surprises. In some cases, access was wider than expected due to overlapping policies. In other cases, legitimate access was blocked because a required relationship was missing or misconfigured.

17.3.2 Identifying gaps and edge cases

A common pattern at ACME was the discovery of edge cases that were technically correct but operationally surprising. For instance, a policy might permit managers of a customer account to access a document, even if they had never directly interacted with that customer. The policy outlined the intended hierarchy accurately, but the runtime behavior raised questions about whether that access was appropriate in all situations.

AI-assisted analysis helped teams quickly identify these scenarios. By clustering decisions and highlighting outliers, AI tools could draw attention to rare permits or unexpected denials. These findings did not automatically indicate a bug. Instead, they prompted a review to see if the original intent still aligned with current expectations.

In several instances, ACME used this insight to improve policies, clarify documentation, or add tests that explicitly covered edge cases revealed by the process. The key point was that runtime analysis guided design-time decisions, completing the loop between intent and behavior.